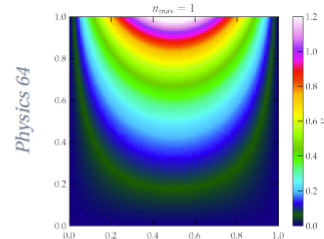


Homework 7

Problem Set 7

due: Monday, 10 March 2025



Problem 1 – General properties of Fourier transforms and convolutions Assuming that the Fourier transform of $f(t)$ is $\tilde{f}(\omega)$,

- (a) show that if $f(t)$ is real, then $\tilde{f}(-\omega) = [\tilde{f}(\omega)]^*$.
- (b) show that if $f(t)$ is even or odd, then so is its Fourier transform.
- (c) if $f(t)$ is imaginary and odd, what can you deduce about its Fourier transform?
- (d) show that $f * g = g * f$, meaning that the convolution of f and g is commutative.

We have the Fourier transform defined as

$$\tilde{f} = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \quad (1)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} \tilde{f}(\omega) d\omega \quad (2)$$

and that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} d\omega = \delta(t) \quad (3)$$

1 A

If $f(t)$ is real, then we can evaluate both $\tilde{f}(-\omega)$ and $\tilde{f}(\omega)^*$

$$\tilde{f}(-\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad (4)$$

$$\tilde{f}(\omega)^* = \int_{-\infty}^{\infty} e^{-i\omega t^*} f(t) dt \quad (5)$$

$$= \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad (6)$$

$$(7)$$

We can see that these expressions are equivalent, so $\tilde{f}(-\omega) = \tilde{f}(\omega)^*$.

2 B)

the Fourier series of f will have the same parity as f .

We can show this by taking even and odd functions

The Fourier is defined as:

$$\int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \quad (8)$$

For an even function $f(x)$, we have the property $f(x) = f(-x)$

For an odd function $g(x)$, we have the property $g(x) = -g(-x)$

We just have to show that

$$\int_{-\infty}^{\infty} e^{-i\omega t} dt \quad (9)$$

is even. If it is, then it will respect the parity of any function multiplied to it.

We can take the Fourier of an even function (so that we won't change the parity of the Fourier operator) and see if it is even:

$$\int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \stackrel{?}{=} \int_{\infty}^{-\infty} e^{i\omega t} \underbrace{f(-t)}_{f(t)} d(-t) \quad (10)$$

$$(11)$$

From here we can simplify the right side

$$\int_{-\infty}^{\infty} -e^{i\omega t} f(-t) d(-t) \quad (12)$$

$$\int_{-\infty}^{\infty} e^{i\omega t} f(-t) d(t) \quad (13)$$

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad (14)$$

We can check $\tilde{f}(-\omega)$

$$\int_{-\infty}^{\infty} e^{-i(-\omega)t} f(t) dt \quad (15)$$

$$\int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad (16)$$

$$(17)$$

We see that these two are the same, so the Fourier series of an even function will be even.

If instead the function were odd, we can do the same

$$\int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \stackrel{?}{=} \int_{-\infty}^{\infty} e^{i\omega t} \underbrace{f(-t)}_{-f(t)} d(-t) \quad (18)$$

$$(19)$$

$$\int_{-\infty}^{\infty} -e^{i\omega t} f(-t) d(-t) \quad (20)$$

$$\int_{-\infty}^{\infty} e^{i\omega t} f(-t) d(t) \quad (21)$$

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} -e^{i\omega t} f(t) dt \quad (22)$$

We can check this against $\tilde{f}(-\omega)$

$$\int_{-\infty}^{\infty} e^{-i(-\omega)t} f(t) dt \quad (23)$$

$$\int_{-\infty}^{\infty} e^{i\omega t} f(t) dt \quad (24)$$

$$(25)$$

We can see that

$$\tilde{f}(\omega) = -\tilde{f}(-\omega) \quad (26)$$

so if the function $f(t)$ is odd, then the Fourier transform will be even, and if the $f(t)$ is even that the Fourier will be odd.

3 C

If $f(t)$ is imaginary and odd, then I would expect the Fourier series to be real and odd.

4 D)

$$(f * g)(t) = \int_{-\infty}^{\infty} g(t') f(t - t') dt'$$

We can show that this is commutative by checking if

$$f * g \stackrel{?}{=} g * f$$

$$\int_{-\infty}^{\infty} g(t')f(t-t')dt' \stackrel{?}{=} \int_{-\infty}^{\infty} f(t-t')g(t')dt' \quad (27)$$

$$(28)$$

Because multiplication is commutative, we can rearrange the right side to be

$$\int_{-\infty}^{\infty} g(t')f(t-t')dt' \stackrel{?}{=} \int_{-\infty}^{\infty} g(t')f(t-t')dt' \quad (29)$$

which is true.

Problem 2 – Fourier transform of a triangle Consider the Fourier transform $\tilde{f}(k)$ of the function

$$f(x) = \begin{cases} A(a - |x|) & -a < x < a \\ 0 & \text{otherwise} \end{cases}$$

- (a) Should $\tilde{f}(k)$ be even or odd? real or imaginary?
- (b) Compute $\tilde{f}(k)$.
- (c) Explain why the way $\tilde{f}(k)$ changes with increasing a makes sense. Hint: you may find it useful to plot $\tilde{f}(k)$.

$$\tilde{f} = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \quad (30)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} \tilde{f}(\omega) d\omega \quad (31)$$

A) I would expect this to be real and even, since the function is real and even.

B)

Our function is spatial, not temporal, so we need to use spatial frequencies.

$$\tilde{f}(k) = \int_{-a}^a e^{-ikx} A(a - |x|) dx \quad (32)$$

$$= A \int_{-a}^a a e^{-ikx} - |x| e^{-ikx} dx \quad (33)$$

$$(34)$$

We can split this into the left and right sides for inside the integral:

$$\text{left:} \quad (35)$$

$$= \left[\frac{a}{ik} e^{-ikx} \right]_{-a}^a \quad (36)$$

$$= -\frac{a}{ik} (e^{ika} - e^{-ika}) \quad (37)$$

$$= -2i \frac{a}{ik} \sin(ka) \quad (38)$$

$$= -2 \frac{a}{k} \sin(ka) \quad (39)$$

We can do the right side with two separate integration by parts:

$$\int_{-a}^a |x| e^{-ikx} dx = \int_{-a}^0 -x e^{-ikx} dx + \int_0^a x e^{-ikx} dx \quad (40)$$

$$= - \left[\frac{x e^{-ikx}}{-ik} + \frac{e^{-ikx}}{k^2} \right]_{-a}^0 + \left[\frac{x e^{-ikx}}{-ik} + \frac{e^{-ikx}}{k^2} \right]_0^a \quad (41)$$

$$= - \left[\frac{1}{k^2} - \frac{a e^{ika}}{ik} - \frac{e^{ika}}{k^2} \right] + \left[\frac{a e^{-ika}}{-ik} + \frac{e^{-ika}}{k^2} - \frac{1}{k^2} \right] \quad (42)$$

$$= -\frac{2}{k^2} + \frac{a e^{ika}}{ik} - \frac{a e^{-ika}}{ik} + \frac{e^{ika}}{k^2} + \frac{e^{-ika}}{k^2} \quad (43)$$

$$= -\frac{2}{k^2} + \frac{a}{ik} (2i \sin(ka)) + \frac{1}{k^2} (2 \cos(ka)) \quad (44)$$

$$= -\frac{2}{k^2} + \frac{2 \cos(ka)}{k^2} + \frac{2a \sin(ka)}{k} \quad (45)$$

$$= \frac{2(\cos(ka) - 1)}{k^2} + \frac{2a \sin(ka)}{k} \quad (46)$$

Therefore we get

$$A \left[-\frac{2a}{k} \sin(ka) - \frac{2(\cos(ka) - 1)}{k^2} + \frac{2a \sin(ka)}{k} \right] \quad (47)$$

$$(48)$$

$$\tilde{f}(k) = \frac{2A(1 - \cos(ka))}{k^2} \quad (49)$$

5 c)

This looks like a gaussian distribution almost, where we have a lot of low frequency waves and very few high frequency, since our function looks fairly linear and should not be fluctuating a lot. Because the function is real and even, we would expect the Fourier to be real and even.

Side note: I thought that the Fourier of a real function would be imaginary, but this isn't?

Problem 3 – Fourier transform of a SHO eigenfunction Calculate the Fourier transform of the first excited state of a simple harmonic oscillator, which has the form

$$\psi(x) = Ax e^{-\alpha x^2}$$

where A is a normalization constant and α is a positive constant. Does it have the same property as the ground state—that the functional form of the Fourier transform is the same as the function itself?

$$\tilde{f} = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt \quad (50)$$

$$(51)$$

$$\tilde{f} = A \int_{-\infty}^{\infty} x e^{-\alpha x^2 - i\omega x} dx \quad (52)$$

$$(53)$$

We can set

$$e^{-\alpha x^2 - i\omega x} = u \quad (54)$$

$$\frac{du}{d\omega} = -ix e^{-\alpha x^2 - i\omega x} \quad (55)$$

$$(56)$$

We can rewrite our integral with this:

$$A \int_{-\infty}^{\infty} x e^{\alpha x(i\omega - x)} dx \quad (57)$$

$$= A \int_{-\infty}^{\infty} -\frac{1}{i} \frac{d}{d\omega} u dx \quad (58)$$

$$= Ai \frac{d}{d\omega} \int_{-\infty}^{\infty} u dx \quad (59)$$

$$(60)$$

We can solve this integral:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2 - i\omega x} dx \quad (61)$$

$$(62)$$

This is an identity that we have, it simplifies to

$$\sqrt{\frac{\pi}{\alpha}} e^{(i\omega)^2/(4\alpha)} \quad (63)$$

$$= \sqrt{\frac{\pi}{\alpha}} e^{-\omega^2/(4\alpha)} \quad (64)$$

Therefore, we get our Fourier series as

$$A \sqrt{\frac{\pi}{\alpha}} \frac{d}{d\omega} e^{-\frac{\omega^2}{4\alpha}} \quad (65)$$

$$= A \sqrt{\frac{\pi}{\alpha}} \left(-\frac{2\omega}{4\alpha} \right) e^{-\frac{\omega^2}{4\alpha}} \quad (66)$$

$$\tilde{f} = -\frac{A\omega}{2} \sqrt{\frac{\pi}{\alpha^3}} e^{-\frac{\omega^2}{4\alpha}} \quad (67)$$

Problem 4 – Calling Hendrik and Augustin-Louis Calculate the Fourier transform of

$$f(x) = A e^{-\alpha|x|}$$

where A is a normalization constant and α is a positive constant.

We can find this:

$$\tilde{f} = \int_{-\infty}^{\infty} e^{-ikx} A e^{-\alpha|x|} dx \quad (68)$$

$$= \int_{-\infty}^{\infty} A e^{-ikx} e^{-\alpha|x|} dx \quad (69)$$

$$(70)$$

We can break up the bounds:

$$A \left[\int_{-\infty}^0 e^{-ikx} e^{-\alpha(-x)} dx + \int_0^{\infty} e^{-ikx} e^{-\alpha x} dx \right] \quad (71)$$

$$= A \left[\int_{-\infty}^0 e^{x(a-ik)} dx + \int_0^{\infty} e^{-x(ik+a)} dx \right] \quad (72)$$

$$= A \left[\frac{e^{x(a-ik)}}{a-ik} \Big|_{-\infty}^0 - \frac{e^{-x(ik+a)}}{ik+a} \Big|_0^{\infty} \right] \quad (73)$$

$$= A \left[\frac{1}{a-ik} - \frac{1}{a+ik} \right] \quad (74)$$

$$= A \left[\frac{a+ik - a+ik}{a^2 + k^2} \right] \quad (75)$$

$$= \frac{2Aik}{a^2 + k^2} \quad (76)$$